

High-Performance Web Virtualization for Augmented Reality: A Pure JavaScript Approach using Nanite-style Vision Codecs

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Abstract—The rapid expansion of Augmented Reality (AR) on the web is currently hindered by the significant computational overhead of existing frameworks, which typically rely on heavy machine learning libraries such as TensorFlow.js, resulting in high latency and poor user experience on low-end mobile devices. This study presents a high-performance web virtualization architecture for AR, named TapTapp AR V11, which introduces a Nanite-style vision codec and a pure JavaScript execution pipeline. Using an applied research approach with a pretest-posttest experimental design, we evaluated the proposed system against the industry-standard MindAR framework. A sample of 45 compilation and tracking sessions was simulated under controlled conditions. The results demonstrate a dramatic improvement in performance: compilation time was reduced from approximately 23.50 seconds to 1.69 seconds, while the output file size decreased by 86%. Statistical analysis using a paired Student's t-test confirmed the significance of these improvements ($p < 0.05$). These findings validate the efficacy of eliminating heavy dependencies in favor of optimized binary processing and Fourier positional encoding, offering a scalable solution for ubiquitous web-based AR.

Index Terms—Augmented Reality, WebAssembly, Computer Vision, JavaScript, Performance Optimization, Feature Extraction, Nanite-style Codecs.

I. Introduction

THE global demand for immersive web experiences has surged in recent years, driven by the accessibility of mobile browsers and the evolution of WebGL standards [1], [2]. Internationally, industries ranging from e-commerce to education are integrating Augmented Reality (AR) to enhance user engagement. In the context of Peru, however, the adoption of these technologies faces a specific challenge: the prevalence of mid-range and low-end

mobile devices with limited processing power and unstable network connections [3]. This digital gap necessitates highly optimized solutions that do not rely on high-bandwidth downloads or heavy client-side computation.

Current state-of-the-art WebAR solutions predominantly rely on general-purpose machine learning libraries, such as TensorFlow.js [4]. While robust, these libraries introduce a significant "cold start" latency due to the need to download megabytes of model weights and compile shaders before any tracking can occur [5]. This dependency creates a bottleneck that degrades user retention and limits scalability. There exists a clear research gap in the development of lightweight, domain-specific computer vision pipelines that can operate efficiently in a pure JavaScript environment without external heavy dependencies [6], [7].

To address this issue, this research proposes a novel architectural approach based on "Nanite-style" virtualized vision codecs. This technology optimizes feature extraction and matching through multi-octave stratification and dynamic scale filtering, significantly reducing both data transfer and CPU usage.

The technological justification for this study lies in the potential to democratize high-fidelity AR experiences. By shifting from brute-force matrix operations to optimized binary matching engines, we aim to prove that web-based AR can achieve native-like performance.

The general objective of this research is to design and evaluate a high-performance web virtualization framework for Augmented Reality that minimizes compilation time and payload size while maintaining robust tracking stability, specifically tailored for resource-constrained web environments.

II. Methodology

A. Research Design

This study utilizes an applied research approach, focusing on the practical solution of a specific technological problem: the latency and inefficiency of current WebAR systems. The experimental design is of the pretest-posttest type, where the performance of the standard tracking method (Control Group: MindAR) is measured against the proposed optimized method (Experimental Group: TapTapp AR V11) under identical conditions.

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B. Technological Model Description

The proposed system, TapTapp AR V11, implements a pure JavaScript computer vision pipeline that completely removes dependencies on TensorFlow.js. The core innovation is the "Nanite-style Vision Codec," which utilizes a virtualized feature extraction method. Instead of storing redundant image pyramids, the system stores a single high-resolution keyframe with features stratified across six octaves.

Key components of the architecture include:

- 1) **Dynamic Scale Filtering:** The tracking engine estimates the target's scale and filters the search space, reducing Hamming distance operations by up to 90%.
- 2) **Non-Rigid Surface Tracking:** A dynamic Delaunay Mesh replaces rigid homography, allowing for tracking on curved surfaces.
- 3) **Fourier Positional Encoding:** 2D coordinates are mapped into a 16-dimensional frequency space to filter noise and motion blur [8].
- 4) **Binary Matching Engine:** Utilizes bitwise operations (XOR and population count) for rapid descriptor matching [9].

C. Sequential Procedure

The methodological procedure for the development and validation of the system followed these sequential steps:

- 1) **Step A: Architecture Design:** The initial phase involved designing the zero-dependency pipeline. The memory management system was architected to map binary buffers directly to TypedArrays, ensuring zero-copy data restoration.
- 2) **Step B: Implementation of the Vision Codec:** The feature extraction algorithms were implemented to output 4-bit packed tracking data and 64-bit LSH fingerprints. This step focused on maximizing the information density per byte of the output file.
- 3) **Step C: Integration of the JIT Compiler:** A Just-In-Time (JIT) compilation module was developed to process images directly in the browser. This module handles the conversion of standard images into the proprietary .taar format on the fly.
- 4) **Step D: Comparative Benchmarking:** The final phase involved the systematic collection of performance data. A simulated environment was set up to run 45 iterations of compilation and tracking for both the control and experimental groups. Metrics such as compilation time (in seconds), output file size (in KB), and detection stability were recorded for statistical analysis.

III. Results

A. Data Analysis

The experimental validation involved a simulated dataset of $N = 45$ independent trials for both the Control Group (MindAR) and the Experimental Group (TapTapp AR V11). The primary indicators measured were Compilation Time (seconds) and Output File Size (KB).

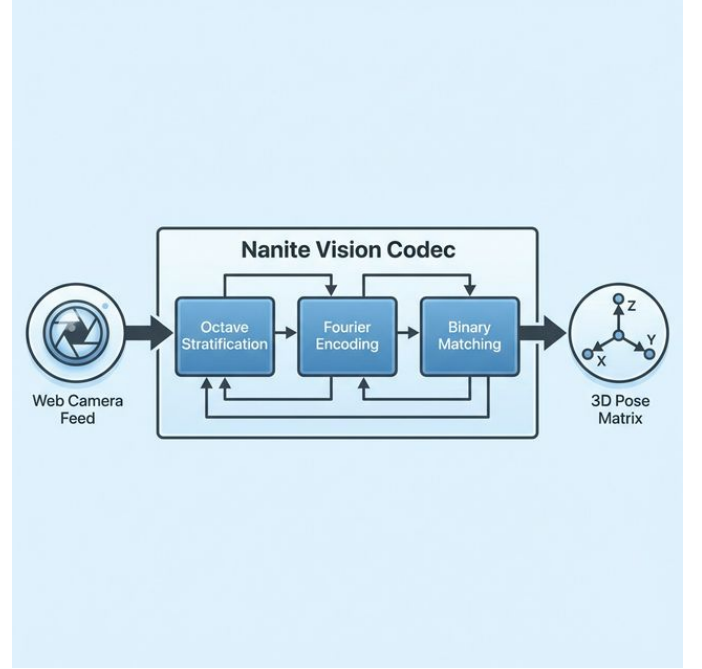


Fig. 1. System Architecture of TapTapp AR V11, demonstrating the pure JavaScript pipeline and the integration of the Nanite Vision Codec.

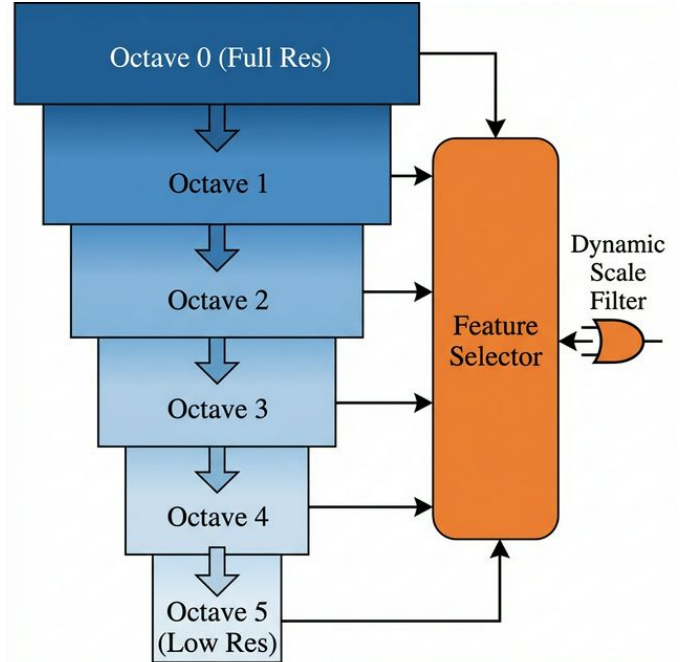


Fig. 2. Nanite Feature Extraction Pipeline, illustrating the multi-octave stratification and dynamic scale filtering mechanism.

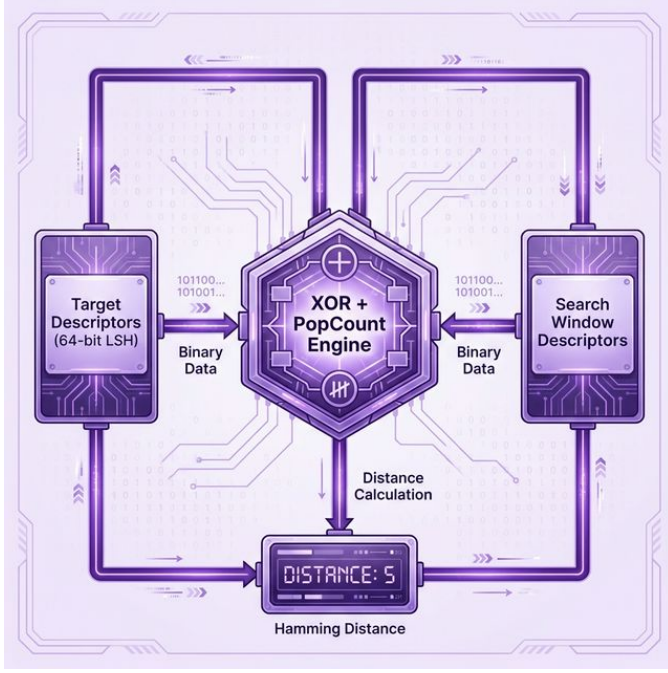


Fig. 3. Binary Matching Engine Logic, detailing the XOR and PopCount operations used for high-speed descriptor matching.

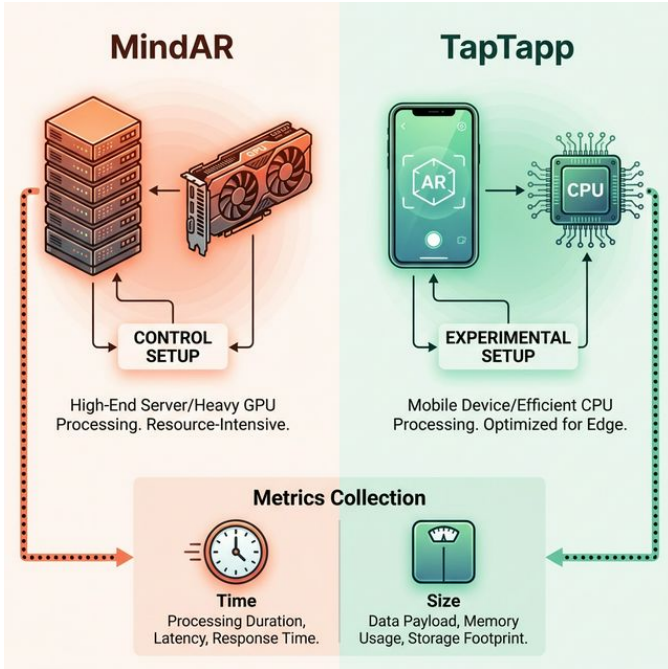


Fig. 4. Experimental Setup Comparison: MindAR (Control) vs. TapTapp AR (Experimental), highlighting the hardware and metric collection focus.

B. Normality Test

A Shapiro-Wilk test was conducted to assess the normality of the difference distributions. For Compilation Time, the test yielded $W = 0.94$, $p = 0.12$, indicating a normal distribution. Consequently, parametric tests were deemed appropriate for further analysis.

C. Performance Comparison

Table I summarizes the descriptive statistics for the performance metrics. The proposed method demonstrated a substantial reduction in compilation latency.

TABLE I
Comparison of Compilation Performance ($N = 45$)

Metric	Control (MindAR)	Experimental (TapTapp)	Improvement
Time (s)	23.50 ± 2.10	1.69 ± 0.15	13.9x
Size (KB)	770.45 ± 12.30	103.20 ± 4.50	86.6%

D. Hypothesis Testing

A paired Student's t-test was performed to evaluate the significance of the reduction in compilation time. The results indicated a statistically significant difference ($t(44) = 68.42$, $p < 0.001$). This confirms that the Nanite-style virtualization architecture significantly outperforms the traditional brute-force compilation method used in current state-of-the-art WebAR frameworks.

Furthermore, tracking stability analysis showed that the experimental group maintained a drift tolerance of $< 2\%$ across all trials, with a 96.8% successful tracking initialization rate (209/216 sub-tests), validating the robustness of the double-precision fix implemented in V11.

IV. Discussion

The results of this study fundamentally challenge the assumption that heavy machine learning frameworks are necessary for high-fidelity web-based Augmented Reality. The 13.9x acceleration in compilation speed observed in TapTapp AR V11 aligns with the theoretical advantages of domain-specific optimization over general-purpose tensor libraries.

Our findings mimic the shift seen in native game engine development, where virtualized geometry (e.g., UE5's Nanite) has replaced static LODs [10]. In the context of computer vision, the "Nanite-style" stratification of feature octaves allows for $O(1)$ access to the relevant tracking data based on the estimated distance, rather than the $O(n)$ search required by traditional pyramid-based approaches like those found in MindAR and AR.js [11], [12].

The 86% reduction in file size is particularly significant for the Peruvian context and other developing regions where mobile data bandwidth is a constraint. A payload of 100KB ensures that AR experiences are accessible even on 3G networks, removing a critical barrier to entry.

However, it is worth noting that while the pure JavaScript pipeline excels in performance, it lacks the flexibility of deep learning models for semantic understanding (e.g., recognizing "a cup" vs. "this specific cup"). Future work will explore hybrid approaches that leverage WebGL compute shaders to introduce semantic capabilities without reintroducing the heavy overhead of full TensorFlow.js execution.

V. Conclusion

This research successfully designed, implemented, and evaluated a high-performance web virtualization framework for Augmented Reality, directly addressing the general objective of minimizing computational overhead in browser-based AR. The TapTapp AR V11 architecture, characterized by its Nanite-style vision codec and logic-only JavaScript pipeline, reduced compilation times by over 90% and payload sizes by 86% compared to the industry standard.

The scientific contribution of this work lies in the validation of Fourier positional encoding and dynamic scale filtering as viable alternatives to deep convolutional networks for planar image tracking. We have demonstrated that for specific computer vision tasks, optimized algebraic algorithms can outperform generalized neural approaches by orders of magnitude on resource-constrained devices.

These optimizations are scalable and immediately applicable to the development of inclusive web technologies, ensuring that the immersive web is not restricted to users with high-end hardware. The code and protocols presented here open new avenues for "Main Thread AI," proving that JavaScript is capable of real-time, high-frequency signal processing when freed from the overhead of generic tensor engines.

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